C. Balkı GEMİRTER

Ph.D. Candidate in Computer Engineering — Yeditepe University



Email cbalkig@gmail.com **Phone** +90 539 293 77 07 **Location** İstanbul, Türkiye

Languages Turkish (Native) | English (Professional) | Japanese (Elementary)

cavidebalki.com ↗ | Google Scholar ↗ | ORCID 0000-0003-3534-3129 ↗ | ResearchGate ↗ | LinkedIn ↗

Deep Learning | Domain Adaptation | Medical Image Analysis | Multimodal Learning | Natural Language Processing (NLP) | Large Language Models (LLM) | Question Answering (MRQA)

6 PUBLICATIONS | **2** THESES | **9** R&D PROJECTS | **19+** YEARS OF EXPERIENCE

RESEARCH PROFILE

Ph.D. candidate in Computer Engineering at Yeditepe University (GPA 4.00/4.00, ranked 1st; expected graduation: December 2026). Doctoral research on incorporating prior knowledge into deep learning architectures for image recognition and medical diagnosis with small cohort data. Author of peer-reviewed publications in SCIE-indexed journals (MDPI Applied Sciences) and IEEE conferences; reviewer for the Turkish Journal of Electrical Engineering & Computer Sciences and the European Journal of Radiology. Guest lecturer at HSE University (Moscow) and for Yeditepe's university-wide multidisciplinary AI 401 course; teaching assistant for Artificial Intelligence, Machine Learning, Deep Learning, and Algorithm Analysis courses. 19+ years of industry R&D leadership supervising TÜBİTAK- and EU-funded projects.

EDUCATION

2020 - Dec 2026 (Expected)

Ph.D. - Computer Engineering

Yeditepe University — Taught in English

4.00 / 4.00 — Ranked 1st, Full Scholarship

THESIS

Incorporating prior knowledge in Deep Learning (DL) architectures for image recognition and medical diagnosis with small cohort data.

2018 - 2020

M.Sc. - Computer Engineering

Yeditepe University — Taught in English

3.87 / 4.00 — Ranked 1st

THESIS

A Turkish Question Answering System Based on Deep Learning Networks (BERT).

2002 - 2007

B.Sc. - Computer Engineering

Ege University

3.34 / 4.00 — Ranked 4th

Education in core computer engineering disciplines and software architecture.

88 ALES 2018 — Academic Personnel and Postgraduate Education Entrance Exam

85 YÖKDİL 2026 — Higher Education Institutions Foreign Language Exam (English)

ACADEMIC PUBLICATIONS

01 **RP-Fuse: Development of an Adaptive Dose-Gated Multimodal Model for Predicting Radiation Pneumonitis in Non-Small Cell Lung Cancer** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, Esra KORKMAZ KIRAKLI, Cemre Ceren ÖZDİLLİ, Ali Kadri ÇIRAK, E. Erkan KORKMAZ, Dionysis GOULARAS | DOI: 10.3390/app162110497

02 **CoDA: A Cognitive-Inspired Approach for Domain Adaptation** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, E. Erkan KORKMAZ, Dionysis GOULARAS | DOI: 10.3390/app16094115

03 **Location Proofing Using Video Similarity** TR Dizin
Journal of Scientific Reports-A (2022) | C. Balkı GEMİRTER, Tacha SERIF

04 **A Turkish Question Answering System Based on Deep Learning Neural Networks** TR Dizin
JISTA - Journal of Intelligent Systems (2021) | C. Balkı GEMİRTER, Dionysis GOULARAS | DOI: 10.38016/jista.815823

05 **Fine-Tuning Throughput and QoS on SMT Cores** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Gürhan KÜÇÜK | DOI: 10.1109/UBMK52708.2021.9559021

06 **A Comparative Evaluation of AMQP, MQTT and HTTP Protocols Using Real-Time Public Smart City Data** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Çağatay ŞENTÜRCA, Şebnem BAYDERE | DOI: 10.1109/UBMK52708.2021.9559032

TEACHING EXPERIENCE

Teaching Assistant · Fall terms · 2021 – Present

Artificial Intelligence CSE 462

Yeditepe University · Dept. of Computer Engineering

Course Instructor: Prof. Dr. E. Erkan Korkmaz — Term-project supervision and grading, assignment grading, and student project consulting. Currently active.

Teaching Assistant · Fall terms · 2021 – Present

Machine Learning CSE 585

Yeditepe University · Dept. of Computer Engineering (Graduate)

Course Instructor: Prof. Dr. E. Erkan Korkmaz — Supervision and grading of graduate term projects and assignment assessment. Currently active.

Teaching Assistant · Fall terms · 2021 – Present

Deep Learning CSE 588

Yeditepe University · Dept. of Computer Engineering (Graduate)

Course Instructor: Prof. Dr. E. Erkan Korkmaz — Supervision and grading of graduate term projects and assignment assessment. Currently active.

Guest Lecturer & TA · Every fall · 2023 – 2026

Artificial Intelligence for Interdisciplinary Studies AI 401

Yeditepe University · University-wide multidisciplinary course

Course Instructor: Prof. Dr. E. Erkan Korkmaz — Delivers a 3-hour lecture every fall semester; designs and grades the assignments.

Guest Lecturer · Spring 2025 & 2026

Artificial Intelligence

HSE University · Moscow, Russia

Course Instructor: Prof. Dr. E. Erkan Korkmaz — Invited 3-hour Artificial Intelligence lectures in two spring terms; designed and assigned the course term projects.

Teaching Assistant · Spring 2022

Algorithm Analysis CSE 311

Yeditepe University · Dept. of Computer Engineering

Course Instructor: Dr. Onur Demir — Course assistantship; assignment and project assessment.

ACADEMIC & PROFESSIONAL SERVICE

Turkish Journal of Electrical Engineering & Computer Sciences

Reviewer · TÜBİTAK

European Journal of Radiology

Reviewer · Elsevier

Bilişim Vadisi (Informatics Valley)

Mentor · Technopark

TÜBİTAK & R&D PROJECTS

TÜBİTAK TEYDEB 1501 · 2026

Image Processing-Supported Patient Monitoring and Fall Risk Management

CEIBA HEALTH

Supervised R&D project focusing on AI-based healthcare solutions.

TÜBİTAK TEYDEB 1832 · 2026

Green and Sustainable Digital Solutions in the Health Sector

CEIBA HEALTH

Supervised R&D initiative for eco-friendly digital health.

TÜBİTAK TEYDEB 1501 · 2024 – 2025

AI-Based Dynamic Customer Segmentation and Personalized Gamification System

Agada Tech

Advanced AI models for user retention and personalized experiences.

TÜBİTAK TEYDEB 1501 · 2023 – 2024

AI-Based Nutrition Analysis System

Vivoo

Multimodal health AI project expanding personalized nutrition insights.

Boğaziçi Technopark · 2022 – 2023

Dragon - Banking Question Answering (MRQA) System

TEB - Türk Ekonomi Bankası

Machine reading comprehension-based smart search over banking documents, giving call center staff instant, concise answers to banking queries.

TÜBİTAK TEYDEB 1501 · 2020 – 2022

AI-Based Customer Chatbot

TEB - Türk Ekonomi Bankası

Developed NEURON, an AI chatbot integrated into digital banking.

TÜBİTAK TEYDEB 1501 · 2016 – 2018

ConnecTEB - VoIP & Multimedia Solution

TEB - Türk Ekonomi Bankası

In-house secure video conferencing tool adopted bank-wide.

TÜBİTAK TEYDEB 1501 · 2014 – 2016

Orion & Aquila (In-House XTM)

TEB - Türk Ekonomi Bankası

Core IT framework development and scaling of ATM/Switch infrastructure.

EU-Supported

InVisIBLe Plus

Yeditepe University

Inclusive and innovative learning tool for visually impaired and blind people.

EXPERIENCE

CEIBA HEALTH · Jan 2026 - Apr 2026

Director of Artificial Intelligence

Bayegan · May 2025 - Jul 2025

Chief AI Officer (CAIO)

Agada Tech · Sep 2024 - Jan 2025

VP of Artificial Intelligence

Yapı Kredi Teknoloji · May 2024 - Aug 2024

AI & Applied Data Science Manager

Vivoo
Chief AI Officer (CAIO) & Data Analytics — Aug 2023 - Feb 2024

TEB - Türk Ekonomi Bankası — Jan 2013 - Aug 2023

- Senior R&D Manager (Artificial Intelligence, Software) — Aug 2016 - Aug 2023
- Infrastructure Software Development Assistant Project Leader — Oct 2015 - Aug 2016
- Switch and ATM Software Development Assistant Project Leader — Nov 2014 - Oct 2015
- Internet Banking Software Development Project Supervisor — Jan 2013 - Nov 2014

NORTEL Networks - NETAŞ — Jul 2007 - Jan 2013

- Senior Software Design Engineer — Mar 2010 - Jan 2013
- Junior Software Design Engineer — Jul 2007 - Mar 2010

CERTIFICATIONS & AWARDS

AWARD — **Leaders of Tomorrow (x2)**
Global Talent Award · 2023 - 2024

Scrum.org — **PSPO-I Professional Scrum Product Owner**
Mar 2020

Scrum.org — **PSM-I Professional Scrum Master**
Feb 2017